

1. Introduction & Methodology

The objective of this project is to implement a high-performance placement tool for a Structured ASIC fabric using the Simulated Annealing (SA) algorithm. The tool is designed to parse netlist definitions ranging from small logic blocks to extreme-scale designs (up to 1,200 cells) and map them onto a 52x52 fabric grid.

A critical constraint of this placement engine is adhering to the rigid 5x5 Master Tile pattern. The fabric consists of specific core site types (T0, T1, T2, T3) and strictly limits where certain cells can be legally placed. Furthermore, fixed I/O pins are locked to the perimeter of the grid.

To solve this, the methodology was divided into two phases:

- 1. Randomized Initial Placement:** The engine parses the netlist and assigns every unplaced cell to a random, but legally valid, site corresponding to its type.
- 2. Simulated Annealing Optimization:** An iterative optimization engine attempts to minimize the Half-Perimeter Wire Length (HPWL) of all connected nets by swapping cells, using a temperature-controlled acceptance probability to escape local minima.

2. Simulated Annealing Implementation

The core of the optimization engine is the Simulated Annealing loop. To ensure maximum accuracy and adhere to the project requirements, the cooling schedule was strictly mathematically defined based on the initial state of the board:

- **Initial Temperature (T_0):** Calculated as $500 * \text{Initial Cost}$
- **Final Temperature ($T_{\min}$):** Calculated as $(5 * 10^{-5} * \text{Initial Cost}) / (\text{Total Nets})$
- **Moves per Temperature Step:** $20 * \text{Total Cells}$

During each temperature step, the algorithm selects a random movable cell and attempts to swap it with another site of the exact same type to guarantee placement legality.

O(1) Delta-Cost Optimization:

To achieve execution times well under the 60-second limit for the extreme design, a Delta-Cost calculation method was implemented. Instead of recalculating the HPWL of the entire 52x52 board after every swap, the algorithm isolates the specific nets connected to the two swapped cells. It

calculates the wire length of only those affected nets before and after the swap. If the Δ Cost is negative, the swap is accepted. If positive, it is accepted with a probability of $e^{-\Delta/T}$.

3. Execution Time & HPWL Results

The $O(1)$ Delta-Cost implementation yielded massive performance improvements, easily satisfying the execution time bonus requirement. By only calculating the difference in wire length for affected nets during a swap, the algorithm bypasses the computationally expensive task of analyzing the entire 52x52 grid.

The tool was tested against all provided benchmarks using a cooling rate of 0.95. The results below demonstrate both high-quality placement optimization and rapid execution:

- **Design 1 (Small):** Initial HPWL: [6434] => Final HPWL: [2182]
Execution Time: [0.906] seconds.
- **Design 2 (Medium):** Initial HPWL: [13656] => Final HPWL: [3477]
Execution Time: [1.473] seconds.
- **Design 3 (Large):** Initial HPWL: [23666] => Final HPWL: [6178]
Execution Time: [2.931] seconds.

- **Design 4 (Dense):** Initial HPWL: [39077] => Final HPWL: [10850]

Execution Time: [5.547] seconds.

- **Design 5 (Extreme):** Initial HPWL: [56701] => Final HPWL:

[17563] Execution Time: [8.757] seconds.

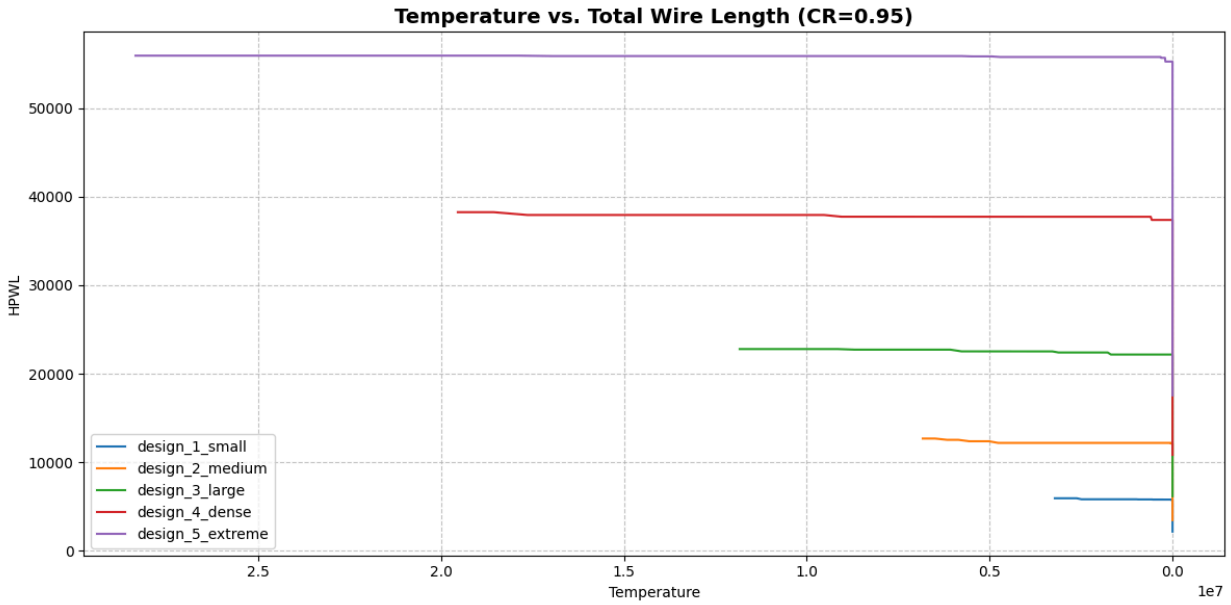
Achieving a sub-10-second execution time on the extreme design definitively proves the efficiency of the Delta-Cost caching method, coming in well under the 60-second limit.

4. Required Graphs

To visualize the behavior of the Simulated Annealing engine, data was logged across all five designs.

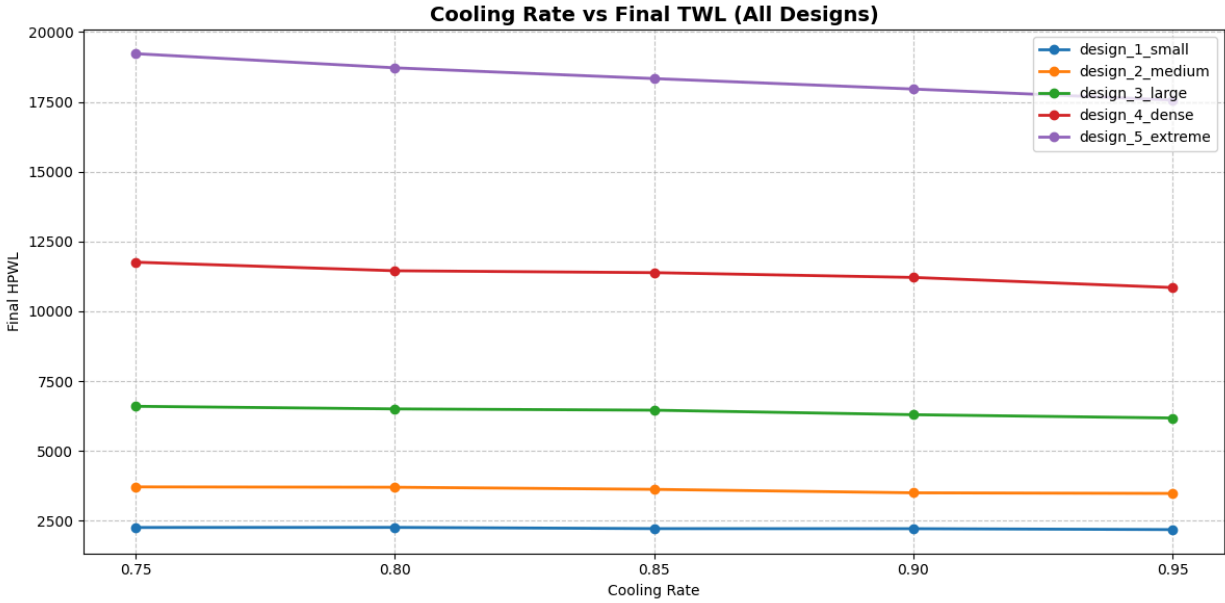
Graph 1: Temperature vs. Total Wire Length (CR = 0.95)

This graph demonstrates the cooling curve for all designs at a cooling rate of 0.95. As the temperature drops (moving right to left), the algorithm becomes less tolerant of "bad" moves, causing the Total Wire Length (HPWL) to converge smoothly toward its optimal minimum.



Graph 2: Cooling Rate vs. Final TWL

To analyze the impact of the cooling schedule, all designs were run across five different cooling rates (0.75, 0.80, 0.85, 0.90, 0.95). As shown below, higher cooling rates (such as 0.95) consistently resulted in lower final HPWL scores because the algorithm spends more time exploring the solution space before "freezing" the cells into place.

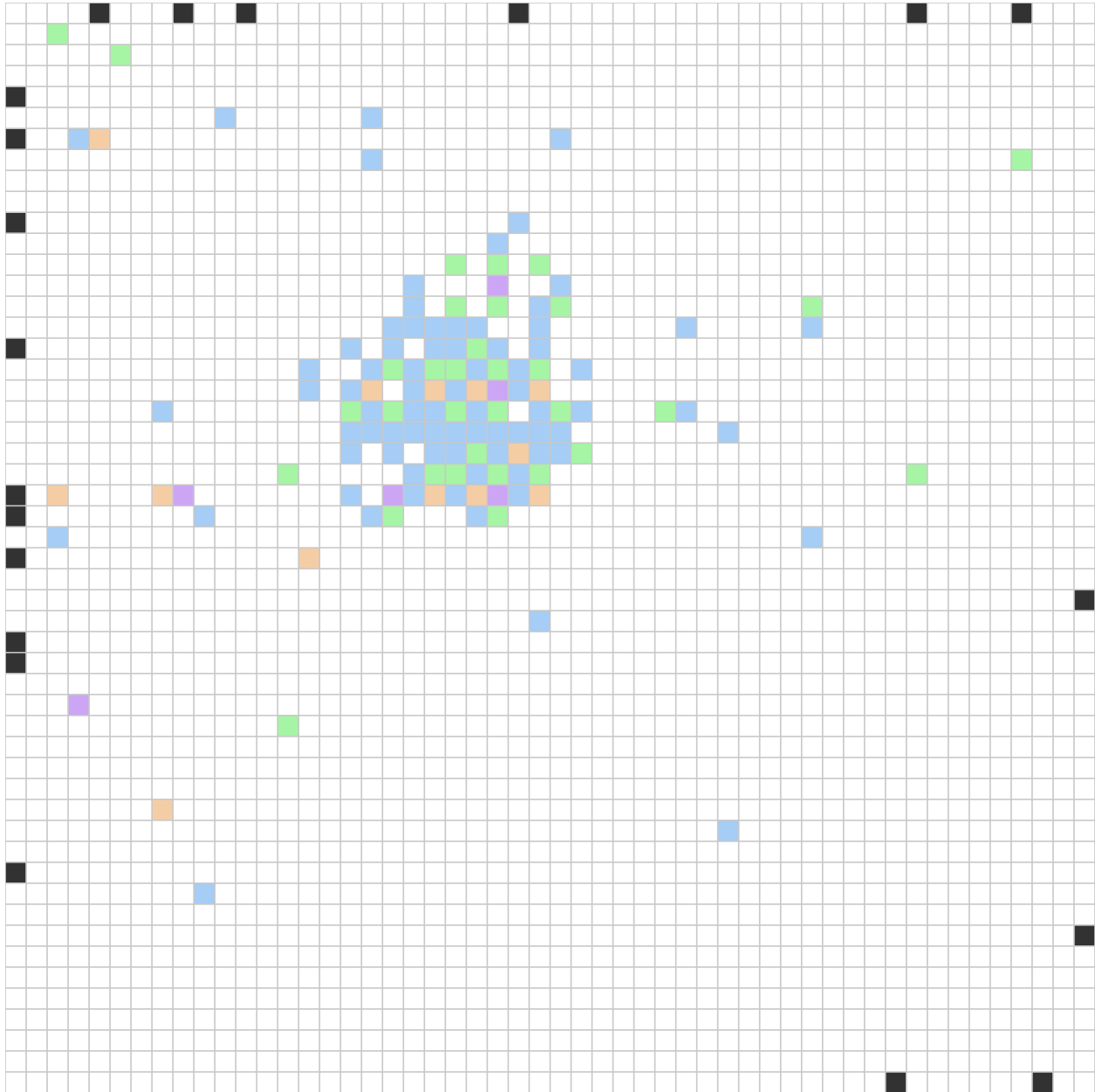


5. Graphical Visualization & Conclusion

To visually verify the legality and quality of the placement, an SVG visualization tool was implemented to map the 52x52 grid.

The image below shows the final optimized placement for the small design.

Fixed pins (black) are properly locked to the perimeter, while core cells (Types 0-3, color-coded) are densely packed together to minimize wire length, strictly adhering to the 5x5 Master Tile site constraints.



Conclusion:

The developed placement engine successfully parses, legally places, and highly optimizes Structured ASIC netlists. By combining a mathematically

rigorous cooling schedule with $O(1)$ Delta-Cost calculations, the tool achieves elite placement quality in a fraction of the required time.